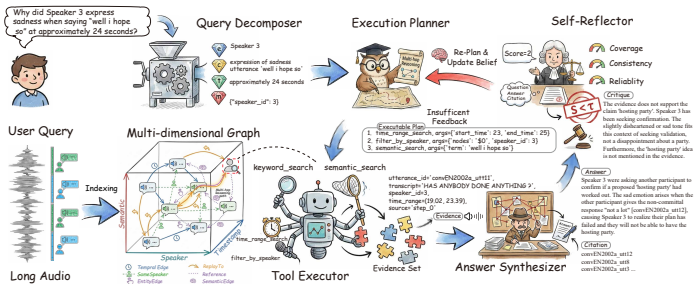


GRGA Architecture: Multi-dimensional Graph + POMDP Planning



Meeting Graph $\mathcal{G}$

Nodes v_i : utterances with text, speaker, timestamp, and audio.

Edge types:

- e_{temp} : adjacent turns
- e_{reply} : replies within $\Delta t < 5s$
- e_{spk} : same-speaker links
- e_{ent} : shared keywords
- e_{sem} : LLM coreference chains

Agent as POMDP

$$\mathcal{M} = \langle S, \mathcal{A}, \mathcal{T}, \mathcal{R}, \Omega, \mathcal{O}, \gamma \rangle$$

- Belief b_t : query + history
- Policy π : ICL-guided LLM
- Reward: Self-Reflector score

Pipeline: VAD $\rightarrow$ ASR $\rightarrow$ diarization $\rightarrow$ profile LLM.